GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M5: Player Client and 3D City

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1. Purpose

M5 is the web page on each player's phone. The player scans a QR code or types the game code, chooses a team, and drives a motorcycle in a low-poly 3D city with buttons on the screen. There is no map on the phone. Only a compass arrow shows the direction of the delivery point or of the store. So players must look at the LED board on the table to find their way.

Runs on	Player phone, only the web browser, nothing to install; the page comes from M2
Technology	TypeScript web page built with Vite; Three.js for 3D; low-poly glTF models from one Blender file, shared with the board buildings; MessagePack; the M3 physics code for prediction
Requirements	R1 R2 R3 R5 R7 R8 R10 R11

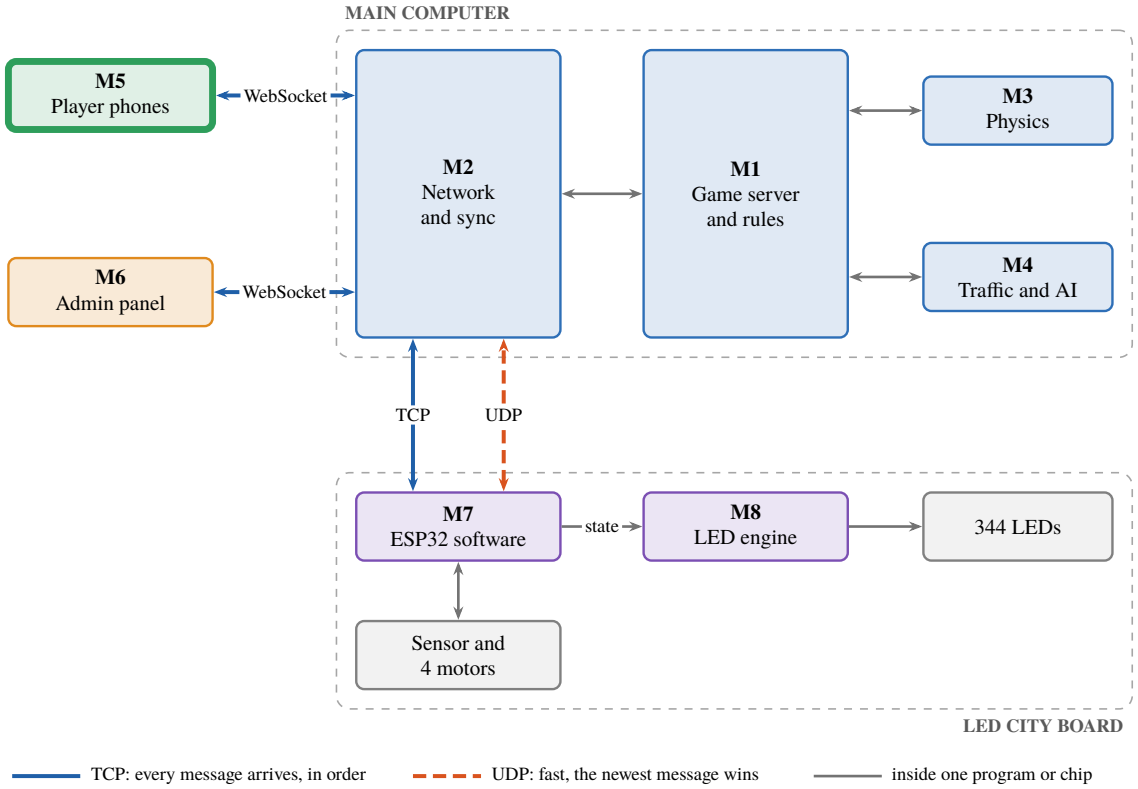


Figure 1: Where M5 sits in the system. Each phone talks to M2 over one WebSocket.

2. Requirements covered

ID	Requirement	What M5 does for it
R1, R2	Game and deliveries	Join, team choice, timer, scores, package icon, "Deliver" button
R3	Driving and controls	Gas, brake, left and right buttons
R5	No map on the phone	Only a compass arrow; "Find me" button

ID	Requirement	What M5 does for it
R7	Earthquake	The camera shakes
R8	Same city as the board	The 3D city is built from the shared city plan and the same models
R10	Network and measurement	Frame-rate limit from the admin; frame-rate report
R11	Single authority	Own motorcycle is predicted at once and corrected by the server

3. Inputs and outputs

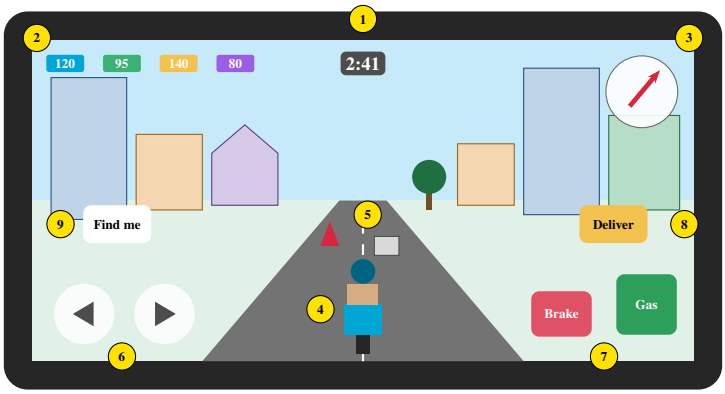
Dir.	Data	From / to	Content
In	Game state	M2	30 times per second: players, NPC cars, obstacles, scores, compass target, “Find me” waiting time
In	Join reply, ping, net config	M2	Player, team, colour tone; server time; frame-rate limit
In	Touch	Player	Buttons
Out	Controls	M2, then M1	On every change and at least 10 times per second: throttle, brake, steer, deliver
Out	Join, “Find me”	M2, then M1	Player key, name, game code, team; “Find me” request
Out	Ping reply	M2	Phone time, frame rate, last state received

4. Main components

Component	What it does	Related modules
Screens and HUD	Join, lobby, countdown, game and end screens; timer, scores, compass	M1
Controls	Turns button presses into the control record	M1, M2
Network client	WebSocket link, MessagePack, reconnect with the saved player key	M2
Prediction	Own motorcycle with the M3 code; correction on each state	M3
3D city and scene	City from the city plan; motorcycles, cars, obstacles, markers; camera	M4, M8

5. How it works

The game screen. The phone is held sideways. The 3D view fills the screen, and the camera follows the motorcycle from behind. At the top the player sees the time left and the score of each team. In the top corner there is the compass. The arrow points to the delivery point, or to the store when the player has no package. It never shows the distance or the roads. The buttons are large so that the thumbs can find them without looking: left and right on one side, gas and brake on the other side. The “Deliver” button is always active. If the server says no, a short message shows the reason, for example “too fast”. The “Find me” button asks the server to flash the player’s LED on the board in white for 1 second. After a press, the button waits for the time that the server sends.



1 Time left
2 Team scores
3 Compass arrow, no map
4 Own motorcycle, team colour
5 NPC car and an obstacle
6 Left and right
7 Gas and brake
8 Deliver button
9 “Find me” button

Figure 2: Planned game screen on a phone held sideways.

Screens. The page has five screens. On the join screen the player types a name and, if there was no QR code, the game code. In the lobby the player chooses one of the four teams and waits. A 3 second countdown starts the game. The game screen is shown above. At the end, the page shows the winner and the scores of all teams.

Smooth play on a real network. The server sends a state 30 times per second, but Wi-Fi adds delay. If the phone waited for the server, the motorcycle would react late to a button. So the phone moves its own motorcycle at once with the M3 code, and corrects it when the real state arrives. For the other players and the NPC cars, the phone keeps the last few states and draws

Figure 3: The five screens of the phone page.

them 100 ms in the past. Then it can always move them smoothly between two real states, and they do not jump.

The 3D city. The city is built from the same city plan as the LED board: 6 roads in each direction, 60 road segments and 25 blocks. Tall buildings stand in the centre, houses and trees are outside, and the four team stores in the corners have the team colours. The models are low-poly and come from one Blender file. The board uses the same models for 3D printing (1 m in the game is 1 mm on the board), so a player can find the same building on the board and on the phone. Objects that repeat, like trees and houses, are drawn as instances, so a mid-range phone can draw at least 30 frames per second. During an earthquake the camera shakes, and the shaking is stronger near the centre of the earthquake.

6. Testing and demonstration

Tests run against a fake server. We test joining on Android and iPhone, every button, prediction with 150 ms delay and 10% loss, at least 30 frames per second on a mid-range phone, and reconnecting after a page reload. In the demonstration two phones drive in the 3D city. The players use the compass and the board to find the delivery point, and the admin changes the frame-rate limit of one phone.